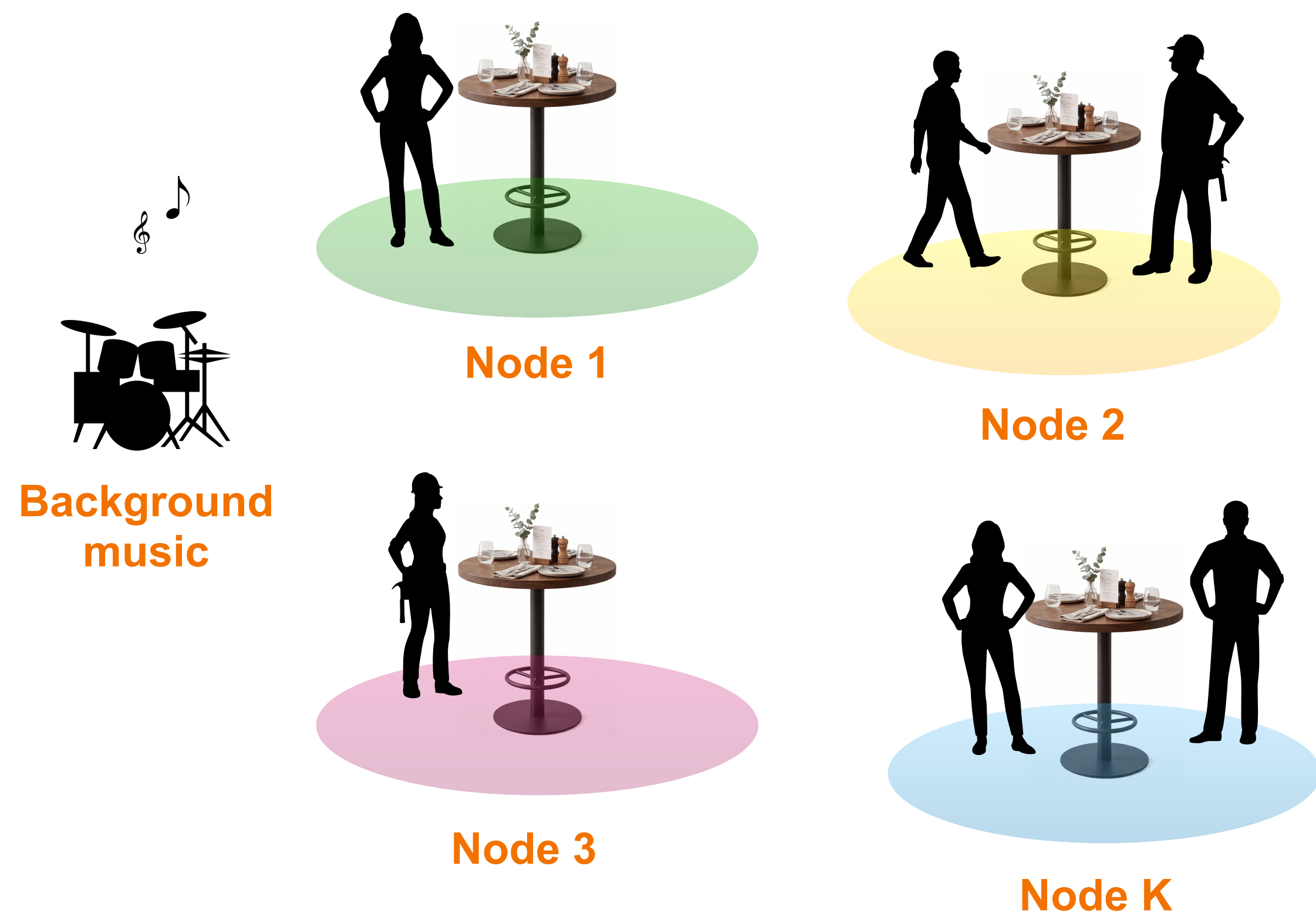


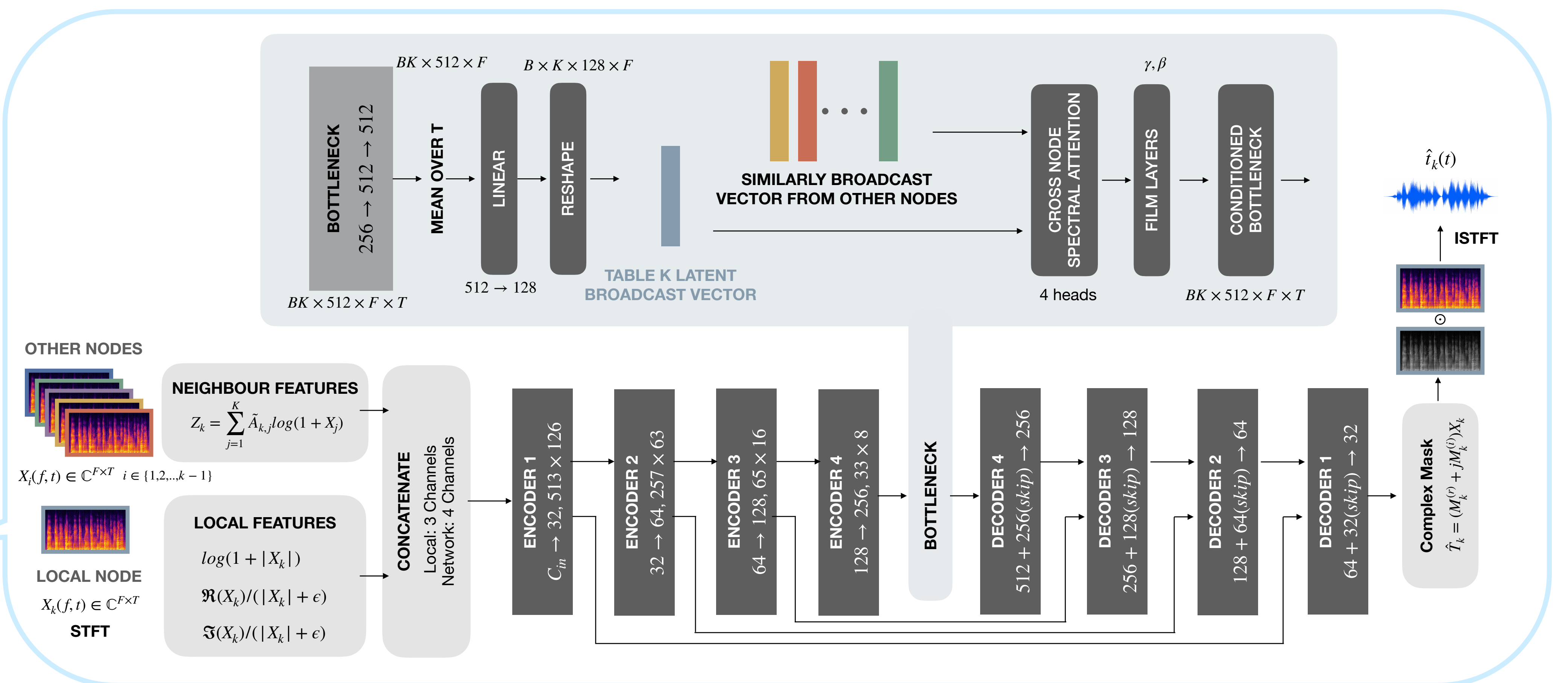
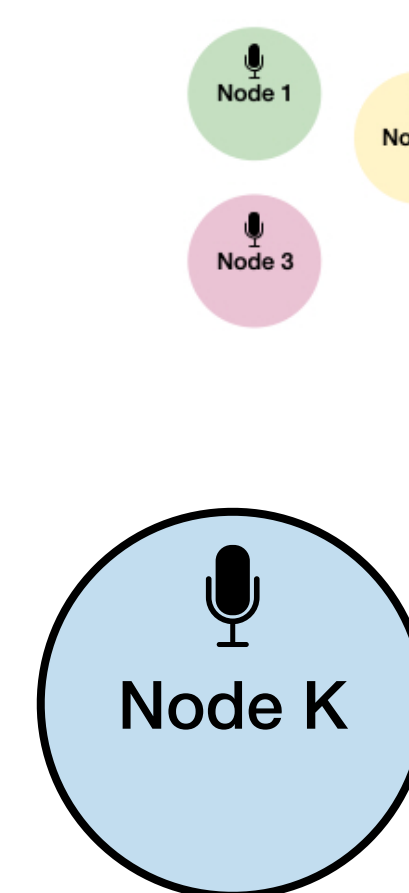
Node-Specific Speech Enhancement in Shared Spaces



- **Each node is single-channel:** one microphone must recover its own local talkers while suppressing competing speech, background music, and noise.
- **Cross-node observations can help,** but transmitting high-dimensional features between nodes consumes communication bandwidth.
- Resource-constrained networks therefore need to balance enhancement performance against communication cost.

Approach

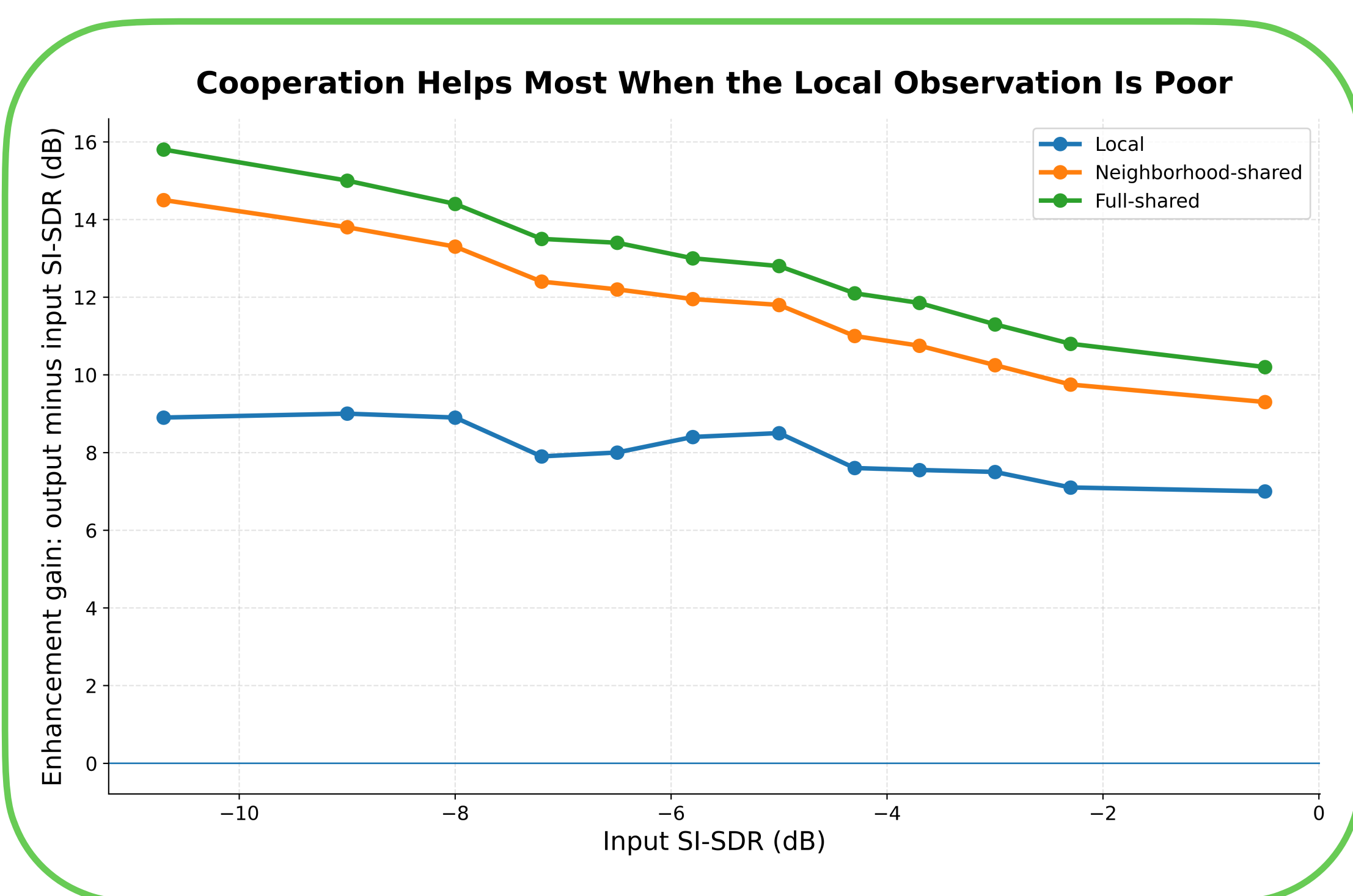
- **Local features:** log-magnitude and normalized complex STFT components from the node's own microphone.
- **Shared features:** log-magnitude information from other nodes; cross-node phase is not exchanged.
- **Graph-conditioned U-Net:** shared information is incorporated at the input and through cross-node attention at the bottleneck to estimate a node-specific mask.



- **Communication Regimes:** Local: own microphone only | Neighborhood: adjacent nodes | Full network: all nodes
- **Why Cooperate?** A single microphone may not contain enough information to separate local speech from competing talkers. Other nodes provide complementary observations of the same acoustic scene.

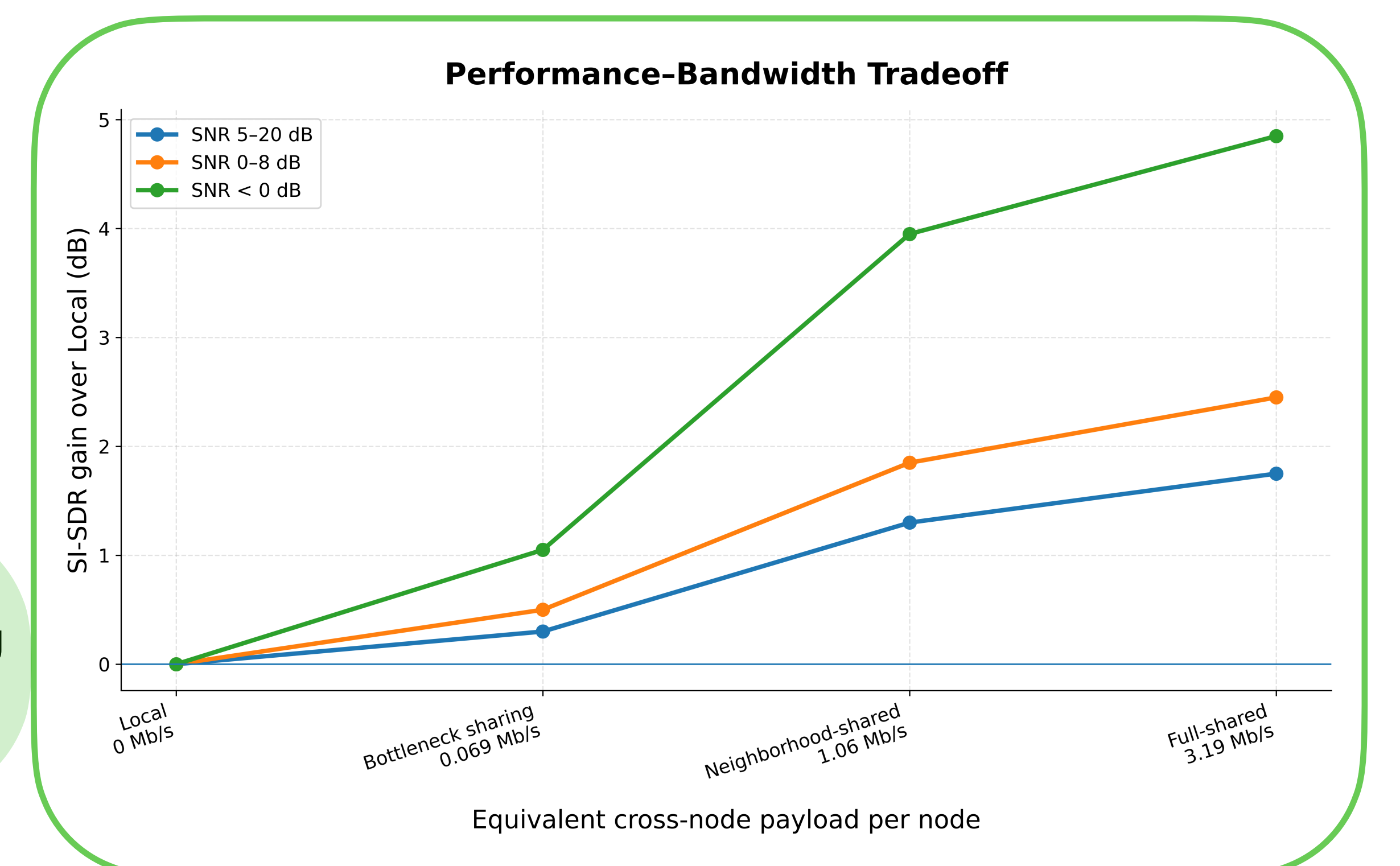
Experiments and Results

- **Experimental Setup:** 4 tables, 1 microphone per table, 5-s scenes, 16 kHz, Speech + music + ambient noise.
- Low / Medium / High acoustic difficulty. **Difficulty increases through lower SNR, more talkers, and stronger reverberation.**

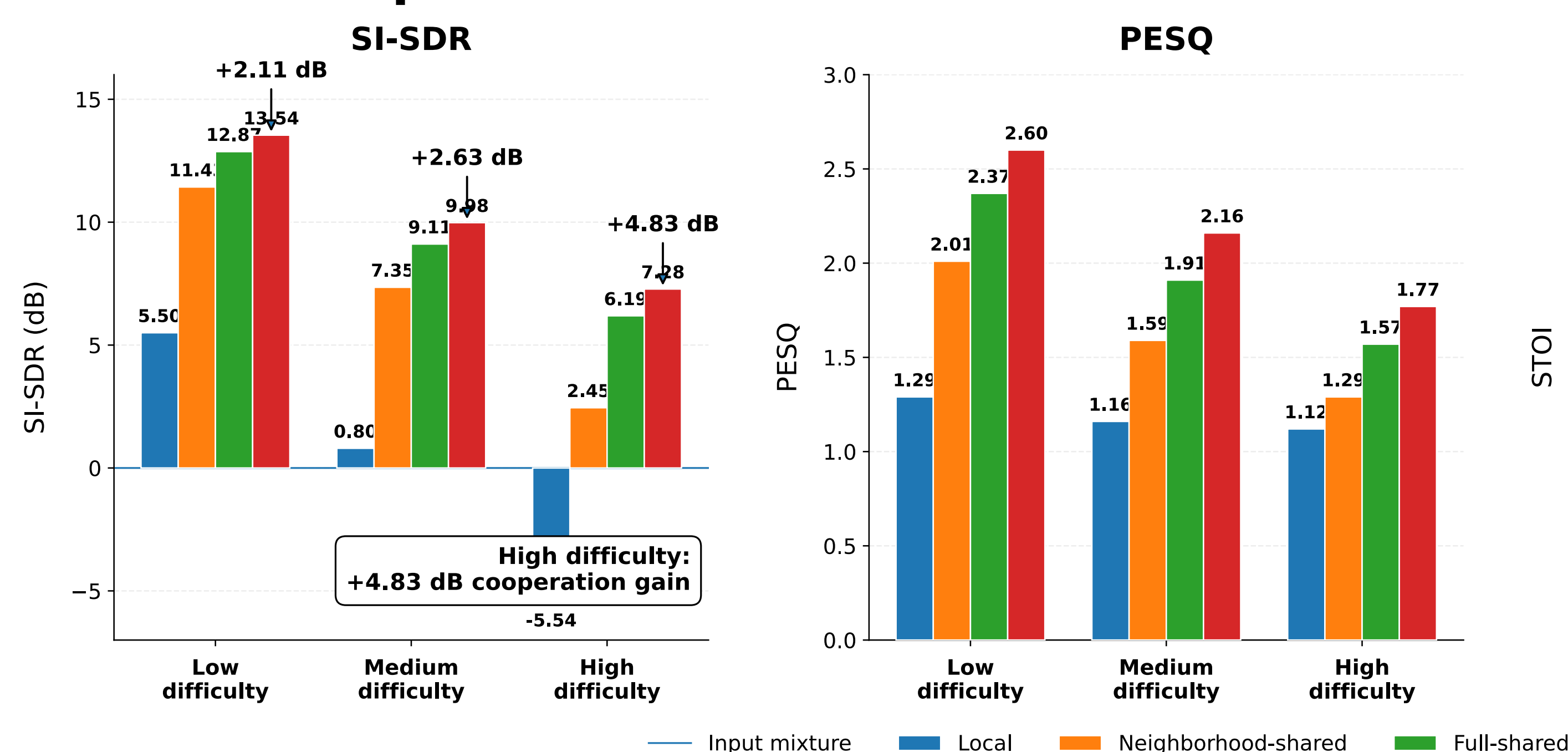


The advantage of cooperative processing increases as input SI-SDR decreases.

Low-rate bottleneck sharing provides modest gains, while richer feature sharing gives larger improvements at higher communication cost.



Cooperation Becomes More Valuable as Conditions Worsen



Difficulty = Low SNR, High RT60, and more speakers. Full-network gain over Local increases from +2.11 dB in low difficulty to **+4.83 dB** in high difficulty. PESQ and STOI show the same ordering.

Learned Models vs. Classical Linear Baselines

